

Lecture 3 Tutorial — Data Distributions

Q1. Key Terms (Fill in the blanks)

Complete the statements.

1. _____ probability refers to the probability of an event given that another event has occurred.
2. _____ probability refers to the probability that two events occur together.
3. A probability distribution can be used as a _____ of uncertainty.

Q2. Variable Type (Multiple Choice)

Choose the correct option.

1. The number of successes in n trials is typically a:
 - A. Continuous variable
 - B. Discrete variable
 - C. Categorical variable
 - D. Joint variable
2. A measurement such as height or time is typically a:
 - A. Discrete variable
 - B. Continuous variable
 - C. Binomial variable
 - D. Poisson variable

Q3. True / False (Probability Reasoning)

Write T (True) or F (False).

1. A low-probability event is impossible.
2. A probability distribution helps describe uncertainty.

Q4. Binomial Distribution Parameters (Matching)

Match each symbol to its meaning.

Symbols:

- A. n
- B. x
- C. p
- D. q

Meanings:

1. Probability of failure in any one trial
2. Number of successes among trials
3. Number of trials
4. Probability of success in any one trial

Q5. Choosing a Distribution (Multiple Choice)

Select the best distribution for each situation.

1. Number of defective items in a sample of 20 items (defective / not defective)
 - A. Binomial
 - B. Poisson
 - C. Gaussian
 - D. Covariance matrix
2. Number of calls received by a help desk in one hour
 - A. Binomial
 - B. Poisson
 - C. Z score
 - D. Joint probability
3. Measurement noise around a mean value (approximately bell-shaped)
 - A. Binomial
 - B. Poisson
 - C. Normal (Gaussian)
 - D. EM

Q6. Short Response (Binomial vs Poisson)

Answer briefly (1–2 sentences each).

1. In the lecture notation, what does μ represent in the Poisson distribution?
2. State one key difference between the role of p in the Binomial distribution and μ in the Poisson distribution.

Q7. Normal Distribution (Fill in the blanks)

Complete the statements.

1. The normal distribution is also called the _____ distribution.
2. The curve is commonly described as _____-shaped and symmetric.
3. In the normal distribution formula, μ typically denotes the _____, and σ denotes the _____.

Q8. Z Score (Formula + Interpretation)

Complete the z-score standardisation formula:

$$x' = \frac{x - \underline{\hspace{2cm}}}{\underline{\hspace{2cm}}}$$

1. If $x > \mu$, is the z score positive, negative, or zero?
2. If $x = \mu$, what is the z score?

Q9. Central Limit Theorem (Fill in the blanks)

Complete the statement from the lecture.

For samples of the same size n , with $n > \underline{\hspace{2cm}}$, the sampling distribution of $\bar{x}$ can be approximated by a normal distribution with mean $\underline{\hspace{2cm}}$ and standard deviation $\underline{\hspace{2cm}}$.

Q10. Central Limit Theorem Key Elements (True / False)

Write T (True) or F.

1. The population must already be normally distributed for CLT to be introduced.
2. The lecture lists “simple random samples” as a key element.
3. The samples should be of the same size n .
4. The population is described using mean μ and standard deviation σ .

Q11. Covariance Matrix and GMM (Multiple Choice + Fill)

1. The lecture introduces Gaussian Mixture Model (GMM) in the context of data distributed in a $\underline{\hspace{2cm}}$ way. (fill in one word)
2. A covariance matrix is primarily used to describe:
 - A. Only the mean of one variable
 - B. Variances and covariances among variables
 - C. Only conditional probability
 - D. Only expected value
3. In a GMM, the mixture weights satisfy:

$$\sum_{i=1}^K \pi_i = \underline{\hspace{2cm}}$$

Q12. Model Fitting, Normality, and EM (Structured Short Answer)

Answer the following concisely.

1. The lecture includes the section titles “Fit a Model” and “Assessing Normality.” Write one question you should ask when evaluating the fitness of a model.
(Use a short, exam-style sentence.)
2. Expand the acronym EM:

$$E = \underline{\hspace{2cm}} \quad M = \underline{\hspace{2cm}}$$